Leave-One-Out Sensitivity Analysis

This supplementary file shows the leave-one-out (loo) sensitivity analysis results for both the frequentist and the Bayesian models.

Method

To assess the influence of individual studies on the pooled effect estimate, we performed a leave-one-study-out (loo) sensitivity analysis under both modeling frameworks. For the frequentist model, we used the `leave1out()` function from `metafor`, which iteratively refits the random-effects model excluding one study at a time. For the Bayesian model, we replicated this procedure manually: the hierarchical model was refit in `brms` twelve times, once excluding each study, using the same model specification, priors, and MCMC settings as the main analysis. For each exclusion, we recorded the resulting pooled effect estimate and its deviation from the estimate obtained using the full dataset.

Frequentist model: code

```
# Frequentist model
freq_model <- readRDS("models/main/freq_full.rds")
loo_freq_results <- leave1out(freq_model)
```

Bayesian model: code

```
# Bayesian model - manual leave1out equivalent
m.brm <- readRDS("models/main/m.brm_full.rds")

priors <- c(
  prior(normal(-1, 2), class = Intercept), # overall effect size μ
  prior(normal(0, 2), class = sd, lb = 0)
```

```r
)

n_studies <- nrow(data_full)

loo_bayes_results <- data.frame(
  excluded_study = character(n_studies),
  estimate = numeric(n_studies),
  lower_ci = numeric(n_studies),
  upper_ci = numeric(n_studies)
)

for (i in 1:n_studies) {
  # remove study i from the data
  data_without_study <- data_full[-i, ]

  # refit the model on the remaining 11 studies
  fit_without_study <- brm(
    beta_ln | se(se_beta_ln) ~ 1 + (1 | author_year),
    data = data_without_study,
    prior = priors,
    save_pars = save_pars(all = TRUE),
    chains = 4,
    iter = 4000
  )

  # Pull out the pooled estimate and CI for this fit
  pooled_estimate <- fixef(fit_without_study)["Intercept", ]

  # Store results in table, row by row
  loo_bayes_results$excluded_study[i] <- data_full$author_year[i]
  loo_bayes_results$estimate[i] <- pooled_estimate["Estimate"]
  loo_bayes_results$lower_ci[i] <- pooled_estimate["Q2.5"]
  loo_bayes_results$upper_ci[i] <- pooled_estimate["Q97.5"]

  # save fits
  saveRDS(
    fit_without_study,
    file = paste0("models/sensitivity/loo/m.brm_excl_", i, ".rds")
  )
}
```

Tab. XX: Loo results frequentist model

excluded_study	estimate	SE	difference_mean
Bauer 2020	-2.706448	0.6686388	-0.25645631
Crump 2013	-2.378536	0.7003712	0.07145566
Dantzer 2020	-2.162945	0.6200586	0.28704672
Desrochers-Couture 2018	-2.756272	0.6437647	-0.30627979
Earl 2016	-2.253747	0.6349425	0.19624540
Hong 2015	-2.420406	0.6876761	0.02958617
Lee 2021	-2.391424	0.6635036	0.05856784
Lucchini 2019	-2.633752	0.6990121	-0.18375970
Lucchini 2012	-2.389250	0.6798677	0.06074204
Menezes-Filho 2018	-2.351857	0.6780156	0.09813536
Roy 2013	-2.280654	0.6524739	0.16933827
Schnaas 2006	-2.726709	0.6553045	-0.27671748

Tab. XX: Loo results Bayesian model

excluded_study	estimate	SE	difference_mean
Bauer 2020	-2.499804	0.6659019	-0.22150856
Crump 2013	-2.174978	0.6683236	0.10331801
Dantzer 2020	-2.034255	0.6362953	0.24404077
Desrochers-Couture 2018	-2.572332	0.6322115	-0.29403640
Earl 2016	-2.115056	0.6432352	0.16323961
Hong 2015	-2.261147	0.6820982	0.01714904
Lee 2021	-2.243753	0.6754269	0.03454269
Lucchini 2019	-2.418512	0.6846228	-0.14021648
Lucchini 2012	-2.209646	0.6744847	0.06864943
Menezes-Filho 2018	-2.161510	0.6625893	0.11678601
Roy 2013	-2.091997	0.6510654	0.18629860
Schnaas 2006	-2.497305	0.6491969	-0.21900957

Results

Excluding individual studies shifted the pooled estimate by up to 0.31 IQ points under the frequentist model and up to 0.29 IQ points under the Bayesian model (column “difference_mean”). The most influential study under both frameworks was Desrochers-Couture 2018, whose exclusion produced the largest deviation from the full-data estimate; this study is notable not for high uncertainty but for very high precision (SE = 0.06), which gives it substantial weight in both models. Averaged across all twelve exclusions, the mean absolute deviation from the full-data estimate was modestly smaller under the Bayesian framework than under the frequentist framework (mean absolute deviation: 0.15 vs. 0.17 IQ points per unit increase in $\log_e$-transformed BLL) though this pattern did not hold for every individual study.